

Molecular energy levels and thermodynamics – summary

Thermodynamics

$$\text{First Law} \quad dU = \delta q + \delta w$$

$$\text{Definition of Helmholtz energy} \quad A = U - TS$$

$$\text{Corresponding Master equation} \quad dA = -p dV - S dT$$

$$\text{hence} \quad \left(\frac{\partial A}{\partial T} \right)_V = -S$$

Definition of partition functions

$$\text{Canonical partition function} \quad Q_N = \sum_{\text{microstates}, m} \exp(-E_m/kT)$$

$$\text{Molecular partition function} \quad q = \sum_{\text{states}, i} \exp(-\varepsilon_i/kT) = \sum_{\text{levels}, j} g_j \exp(-\varepsilon_j/kT)$$

$$\text{separation} \quad q = q_{\text{trans}} q_{\text{rot}} q_{\text{vib}} q_{\text{elec}}$$

$$\text{Indistinguishable particles} \quad Q_N = \frac{q^N}{N!} \quad N! \text{ goes with } q_{\text{trans}}$$

Relation of thermodynamic parameters to partition functions

$$\text{Bridge relationship} \quad A = -kT \ln Q_N$$

$$\text{via } (\partial A / \partial T)_V = -S \quad S = k \ln Q_N + kT \left(\frac{\partial \ln Q_N}{\partial T} \right)_V$$

$$\text{via } A = U - TS \quad U = kT^2 \left(\frac{\partial \ln Q_N}{\partial T} \right)_V$$

In terms of the molecular partition function

$$\text{for all modes} \quad U = NkT^2 \left(\frac{\partial \ln q}{\partial T} \right)_V$$

$$S_{\text{trans}} = Nk \left(\ln q_{\text{trans}} - \ln N + 1 + T \left(\frac{\partial \ln q_{\text{trans}}}{\partial T} \right)_V \right) = Nk (\ln q_{\text{trans}} - \ln N + 1) + \frac{U_{\text{trans}}}{T}$$

$$\text{for rotation, vibration etc.} \quad S_{\text{rot}} = Nk \left(\ln q_{\text{rot}} + T \left(\frac{\partial \ln q_{\text{rot}}}{\partial T} \right)_V \right) = Nk \ln q_{\text{rot}} + \frac{U_{\text{rot}}}{T}$$

Equipartition principle

If the spacing of the energy levels is $\ll kT$, each ‘degree of freedom’ contributes $\frac{1}{2}NkT$ to the internal energy. In molar units, the contribution is $\frac{1}{2}RT$.

Partition functions

$$\text{Translation} \quad E_n = \frac{n^2 h^2}{8ma^2} \quad n = 1, 2, \dots \quad q_{\text{trans,3D}} = \left(\frac{2\pi mkT}{h^2} \right)^{\frac{3}{2}} V$$

$$\text{Rotation} \quad E_J = BJ(J+1) \quad J = 0, 1, \dots$$

$$B = \frac{\hbar^2}{2I} \quad \tilde{B} = \frac{h}{8\pi^2 c I} \quad \text{diatomic } I = \mu R^2 \quad \mu = \frac{m_1 m_2}{m_1 + m_2}$$

$$\text{if } kT \gg B \quad q_{\text{rot}} = \frac{kT}{\sigma B} = \frac{T}{\sigma \theta_{\text{rot}}} \quad \theta_{\text{rot}} = \frac{B}{k}$$

$\sigma = 1$ for heteronuclear and $= 2$ for homonuclear diatomics.

Vibration – measuring from the bottom of the PE well

$$E_v = (v + \frac{1}{2})h\nu_0 \quad v = 0, 1, \dots \quad q_{\text{vib}} = \frac{\exp(-h\nu_0/2kT)}{1 - \exp(-h\nu_0/kT)} = \frac{\exp(-\theta_{\text{vib}}/2T)}{1 - \exp(-\theta_{\text{vib}}/T)} \quad \theta_{\text{vib}} = \frac{h\nu_0}{k}$$

Vibration – measuring from the ground state

$$q'_{\text{vib}} = \frac{1}{1 - \exp(-h\nu_0/kT)} = \frac{1}{1 - \exp(-\theta_{\text{vib}}/T)}$$

Nuclear spin statistics

On swapping identical nuclei: Bosons $\Psi \rightarrow +\Psi$; Fermions $\Psi \rightarrow -\Psi$

Symmetry operations to swap nuclei in a homonuclear diatomic: C_2 i_{elec} σ_{elec} p_{nuc}

Number of symmetric nuclear spin states: $(I+1)(2I+1)$,

Number of anti-symmetric nuclear spin states: $I(2I+1)$.

Nuclei with $I = 0, 1, \dots$ are Bosons; those with $I = \frac{1}{2}, \frac{3}{2}, \dots$ are Fermions.

Equilibrium constants

$$\mu_i = -kT \ln \frac{q_i}{N_i} + \varepsilon^{0,i}$$

For the general reaction $\nu_A A + \nu_B B \rightleftharpoons \nu_M M + \nu_N N$

$$K_c = \frac{f_M^{\nu_M} f_N^{\nu_N}}{f_A^{\nu_A} f_B^{\nu_B}} \left(\frac{1}{c^\circ} \right)^{\Delta \nu} \exp(-\Delta \varepsilon_0/kT)$$

$$\Delta \varepsilon_0 = \nu_M \varepsilon^{0,M} + \nu_N \varepsilon^{0,N} - \nu_A \varepsilon^{0,A} - \nu_B \varepsilon^{0,B} \quad \Delta \nu = \nu_M + \nu_N - \nu_A - \nu_B$$

where f is the volume independent partition function.

$$K_p = \frac{(q_M^\circ)^{\nu_M} (q_N^\circ)^{\nu_N}}{(q_A^\circ)^{\nu_A} (q_B^\circ)^{\nu_B}} (N_A)^{-\Delta \nu} \exp\left(-\frac{\Delta \varepsilon_0}{kT}\right)$$

where q° is the partition function in the standard state (i.e. the volume is the standard molar volume)

Rate constants

For A + BC:

$$k_{\text{2nd}} = \frac{kT}{h} \frac{f'_{\text{TS}}}{f_{\text{A}}f_{\text{BC}}} \exp(-\Delta\varepsilon_0^\ddagger/kT)$$

Thermodynamic formulation (solution) $k_{\text{2nd}} = \frac{1}{[\text{ }]^\circ} \frac{kT}{h} \exp\left(\frac{\Delta_{\text{r}}S^{\circ,\ddagger}}{R}\right) \exp\left(\frac{-\Delta_{\text{r}}H^{\circ,\ddagger}}{RT}\right)$

Boltzmann distribution

$$n_i = \frac{N}{q} \exp\left(\frac{-\varepsilon_i}{kT}\right) \quad n_j = \frac{N}{q} g_j \exp\left(\frac{-\varepsilon_j}{kT}\right)$$